

Ch-2.

EXERCISE

pic.

1. ⁸⁶ State and explain parallel axis theorem
2. ⁸⁷ State perpendicular axis theorem.
3. Define :
 - (i) Moment of inertia ⁷⁹
 - (ii) Section modulus ⁸²
 - (iii) Radius of gyration ⁸³

(Nov. 2018, 2019, May 2019, Feb. 2021, Sept. 2021, Feb. 2022)

(Nov. 2016, May 2019, Feb. 2021, 2022)

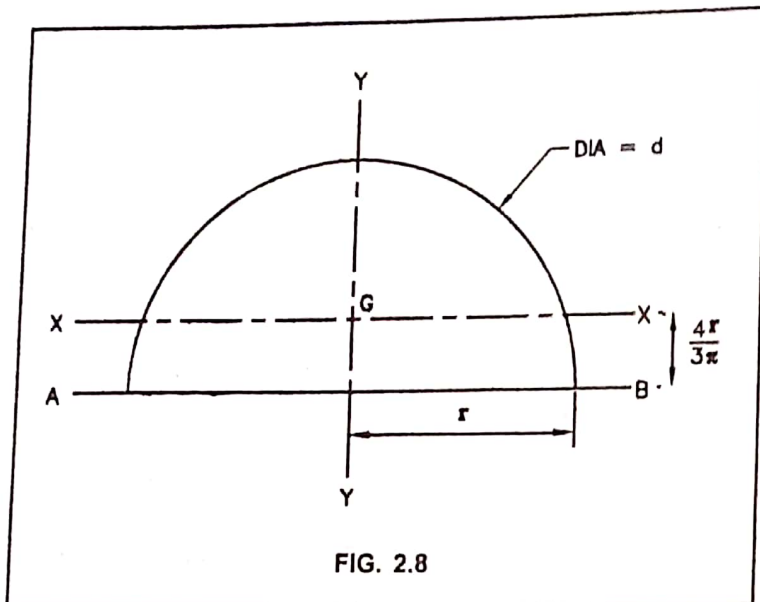
(May 2016, 2017, 2019, Nov. 2019, Sept. 2021)

(Nov. 2016, 2018)

(Nov. 2016, 2017, 2018, May 2019, Sept. 2021)



(6) Semicircular section :



$$I_{xx} = 0.11 r^4$$

$$I_{AB} = \frac{\frac{\pi}{64} d^4}{2} = \frac{\pi}{128} d^4$$

$$I_{yy} = \frac{\pi}{8} r^4$$

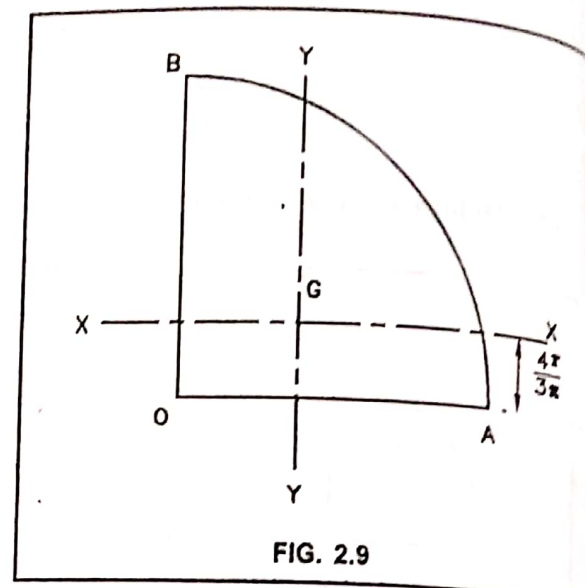
$$= 0.393 r^4$$

$$= \frac{\pi}{128} d^4$$

(7) Quarter circle :

$$I_{xx} = I_{yy} = 0.055 r^4$$

$$I_{OA} = I_{OB} = \frac{\pi}{16} r^4$$



2.5 PARALLEL AXIS THEOREM :

"If I_g is the moment of inertia of a plane area about an axis passing through its centre of gravity, then, moment of inertia of the area about axis AB, parallel to the first axis, and at a distance h from centre of gravity is given by

$$I_{AB} = I_g + ah^2$$

Where, I_{AB} = M.I. of area about AB

I_g = M.I. of area about c.g.

a = area of the section

h = distance between c.g. of section and axis AB.

Proof : Consider a small elemental strip of area da ,

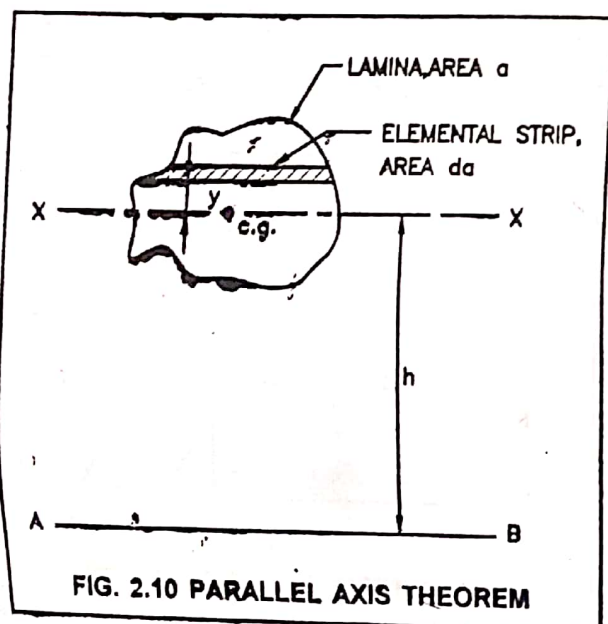
y = distance of strip from c.g. of section

$\therefore$ M.I. of strip about c.g. of section,

$$\therefore I = da \cdot y^2$$

$\therefore$ M.I. of whole section about an axis passing through c.g.

$$I_g = \sum da \cdot y^2$$



∴ M.I. of whole section about the axis AB,

$$\begin{aligned} I_{AB} &= \Sigma da \cdot (h + y)^2 \\ &= \Sigma da \cdot (h^2 + 2hy + y^2) \\ &= \Sigma h^2 \cdot da + \Sigma 2hy \cdot da + \Sigma y^2 \cdot da \\ &= ah^2 + 0 + I_g \end{aligned}$$

$$\therefore \boxed{I_{AB} = I_g + ah^2}$$

$$\therefore \Sigma h^2 \cdot da = ah^2$$

$$\Sigma y^2 \cdot da = I_g$$

$$\Sigma da \cdot y = a \bar{y} = 0$$

$$\text{because } \bar{y} = 0$$

2.6 PERPENDICULAR AXIS THEOREM : 02

(Nov. 2018, May 2019, Feb. 2021, 2022)

"If, I_{xx} and I_{yy} be the moments of inertia of a plane section about two perpendicular axes meeting at O , the moment of inertia I_{zz} about the $z - z$ axis, perpendicular to the plane and passing through the intersection of $x - x$ and $y - y$ axes is given by :

$$I_{zz} = I_{xx} + I_{yy}$$

Proof :

Consider a small lamina (P) of area da .

x and y are the co-ordinates of P along OX and OY axes.

$$OP = r$$

$$\therefore r^2 = x^2 + y^2$$

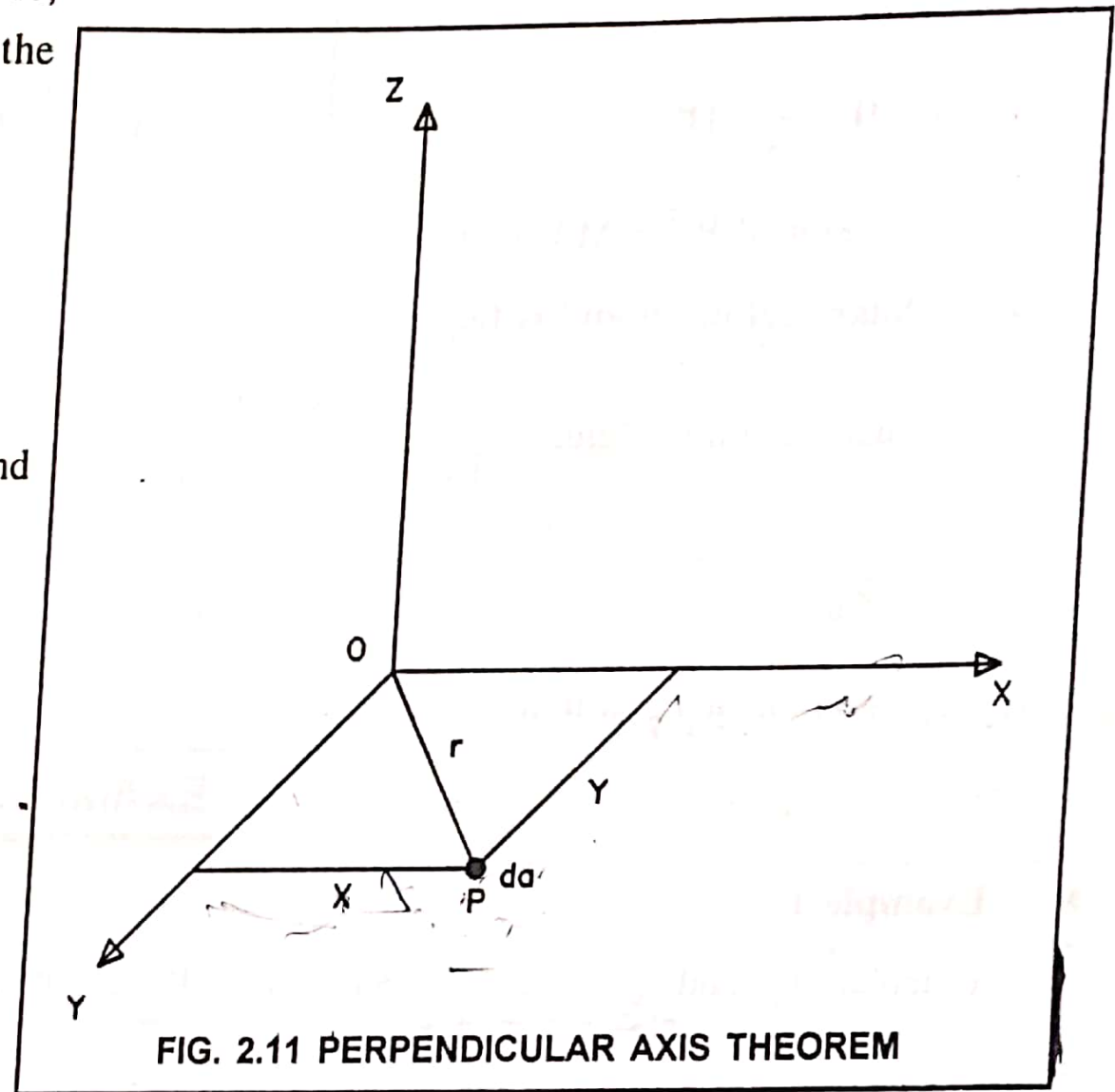
M.I. of P about $x - x$ axis,

$$I_{xx} = da \cdot y^2$$

similarly, $I_{yy} = da \cdot x^2$

$$I_{zz} = da \cdot r^2 = da (x^2 + y^2) = da \cdot x^2 + da \cdot y^2$$

$$\therefore I_{zz} = I_{xx} + I_{yy}$$



2.1 **MOMENT OF INERTIA (I) :** 03 (P)

(May 2016, 2017, 2019, Nov. 2019)

The second moment of area about any axis is called **moment of inertia**.

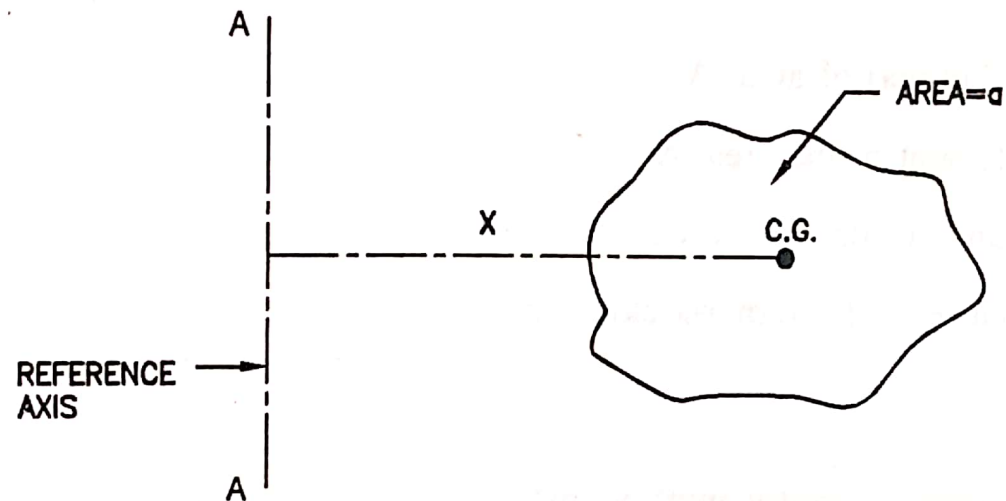


FIG. 2.1 MOMENT OF INERTIA

The First moment of area = $a \cdot x$

The second moment of area = $(ax)x$

$$= ax^2 = \text{M.I.}$$

$\therefore$ Moment of inertia = $I = ax^2$

The unit of M.I. is mm^4 , cm^4 or m^4 .

2.2 SECTION MODULUS : (Z) 03(ii)

(May 2015, Nov. 2016, 2018, Sept. 2021)

Section modulus = $\frac{\text{M.I.}}{\text{distance of extreme fibre from neutral axis}}$

$$\therefore \boxed{Z = \frac{I}{y_{\max}}}$$

unit of Z is mm^3 or cm^3 .

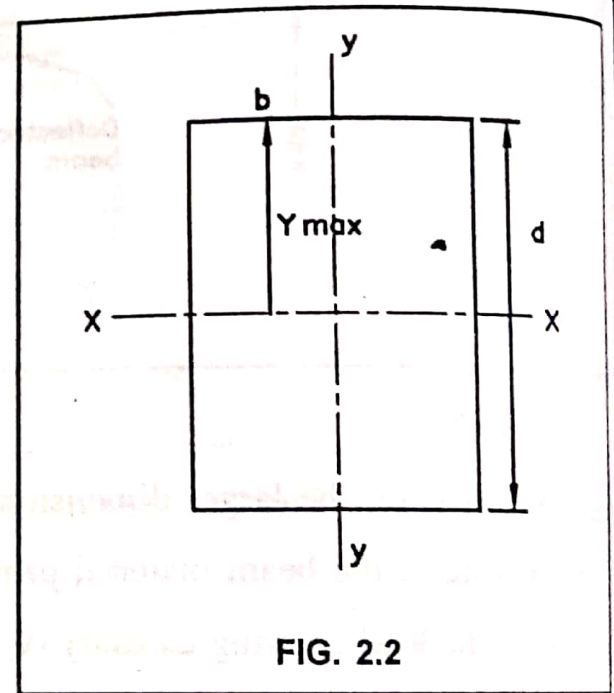
For rectangular section,

$$Z_{xx} = \frac{I_{xx}}{y} = \frac{I_{xx}}{\frac{d}{2}}$$

$$Z_{yy} = \frac{I_{yy}}{x} = \frac{I_{yy}}{\frac{b}{2}}$$

Section modulus is the first moment of area.

$$\therefore Z = \int x \cdot dA \text{ or } = \int y \cdot dA$$



2.3 RADIUS OF GYRATION (k):

(Nov. 2016, 2017, 2018, May 2019, Sept. 2021)

The distance from the given axis, at which, if all the small elements of the given lamina are placed, the M.I. of the lamina about the given axis does not change. This distance is called **radius of gyration**.

Mathematically,

$$I = A k^2$$

$$\therefore k = \sqrt{\frac{I}{A}}$$

Where, I = moment of inertia

A = Area

k = radius of gyration.

Fig. 2.2 (a) shows a lamina of area A.

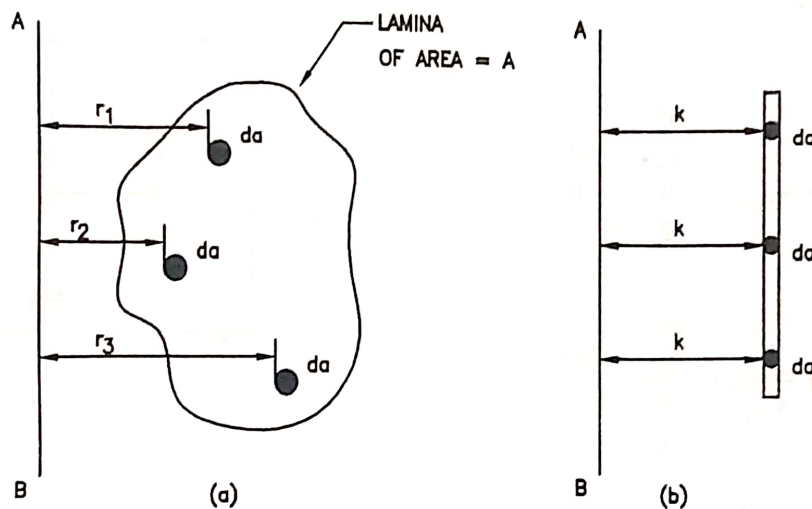


FIG. 2.2 (a) RADIUS OF GYRATION

Lamina is split up into small elements each of area da .

Let $r_1, r_2, r_3 \dots$ etc. are distances of small elements from axis AB.

$$\therefore I_{AB} = da \cdot r_1^2 + da \cdot r_2^2 + da \cdot r_3^2 + \dots \quad \left| \quad \Sigma da = A \right.$$

$$\therefore I_{AB} = \Sigma da \cdot r^2$$

Let, the individual small elements of lamina be arranged at the same distance k from axis AB, the distance k being such that the moment of inertia about the axis AB remains unchanged.

In this second position, each small element area is at the same distance k from AB.

$\therefore$ M.I. of the lamina @ axis AB

$$I_{ab} = da \cdot k^2 + da \cdot k^2 + da \cdot k^2 + \dots$$

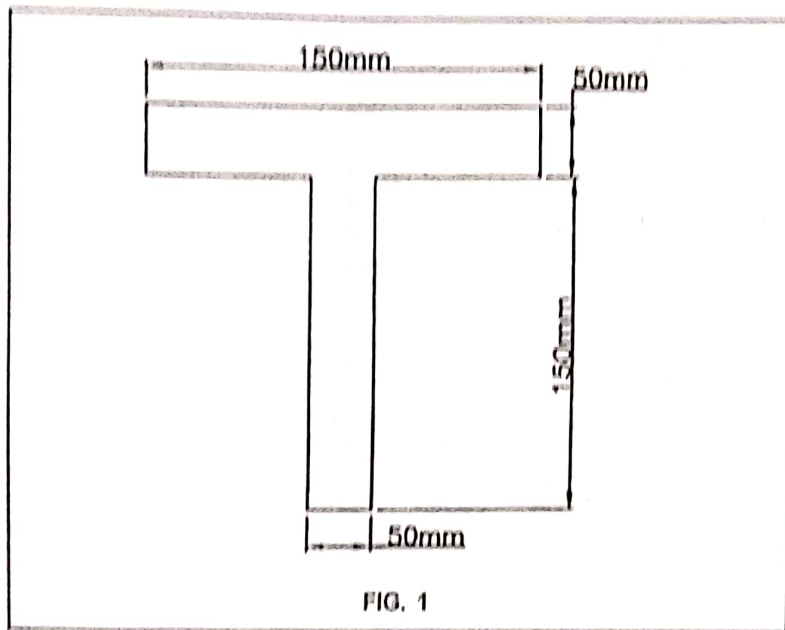
$$= \Sigma da \cdot k^2$$

$$= A \cdot k^2$$

k is called radius of gyration of lamina about axis AB.

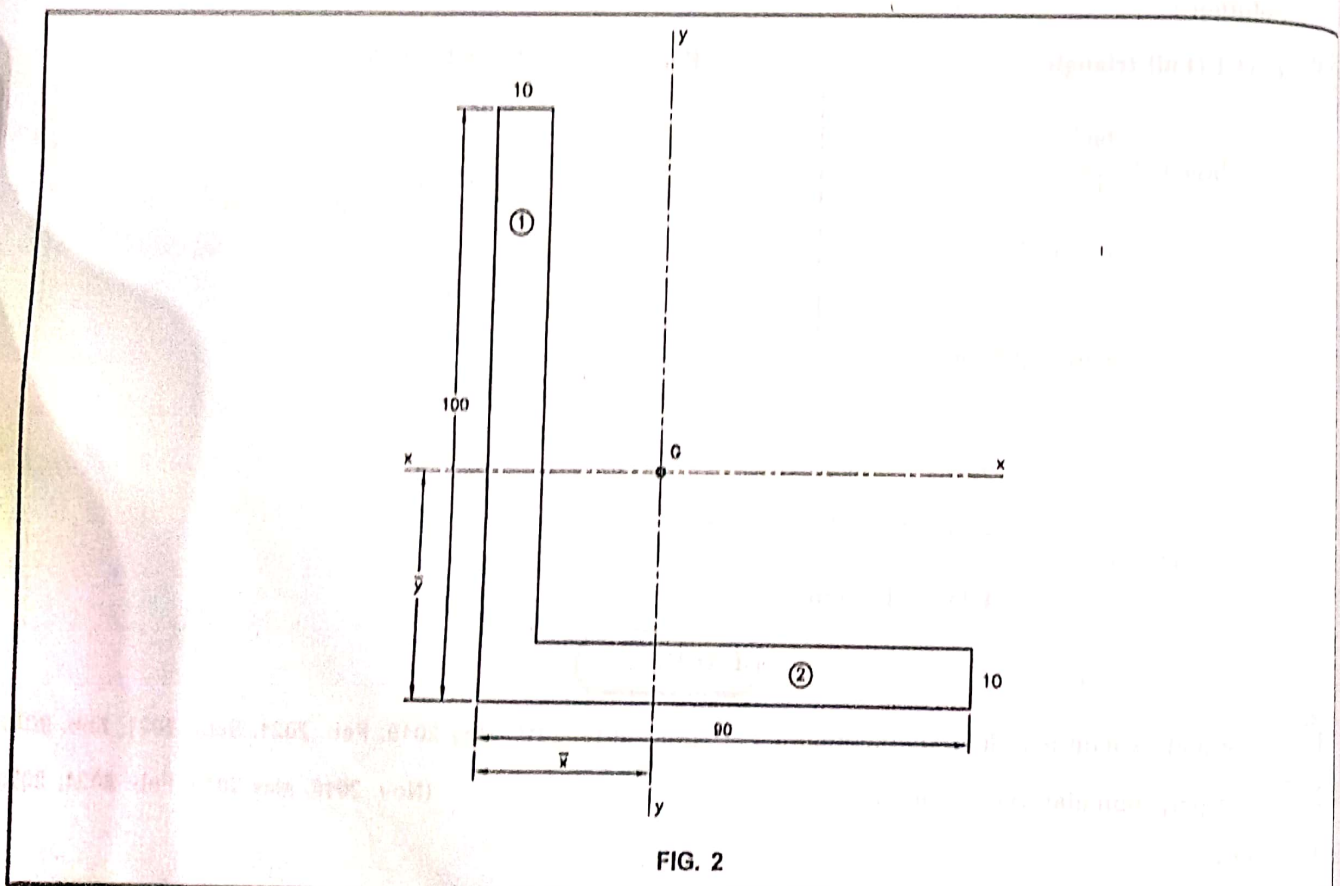
Ch-2.

EXERCISE - NUMERICALS

Ex. 1 Find I_{xx} and I_{yy} for a T-section given below.[Ans. : $\bar{x} = 75 \text{ mm}$, $\bar{y} = 125 \text{ mm}$

$$I_{xx} = 53.125 \times 10^6 \text{ mm}^4$$

$$I_{yy} = 15.625 \times 10^6 \text{ mm}^4]$$

Ex. 2 Find M. I. I_{xx} and I_{yy} for Angle section $100 \times 90 \times 10 \text{ mm}$ shown in fig.[Ans. : $\bar{x} = 25 \text{ mm}$, $\bar{y} = 30 \text{ mm}$

$$I_{xx} = 1.74 \times 10^6 \text{ mm}^4$$

$$I_{yy} = 1.33 \times 10^6 \text{ mm}^4]$$

Ex. 3 Find M. I. at the base of an ISA 90 × 90 × 10 mm.

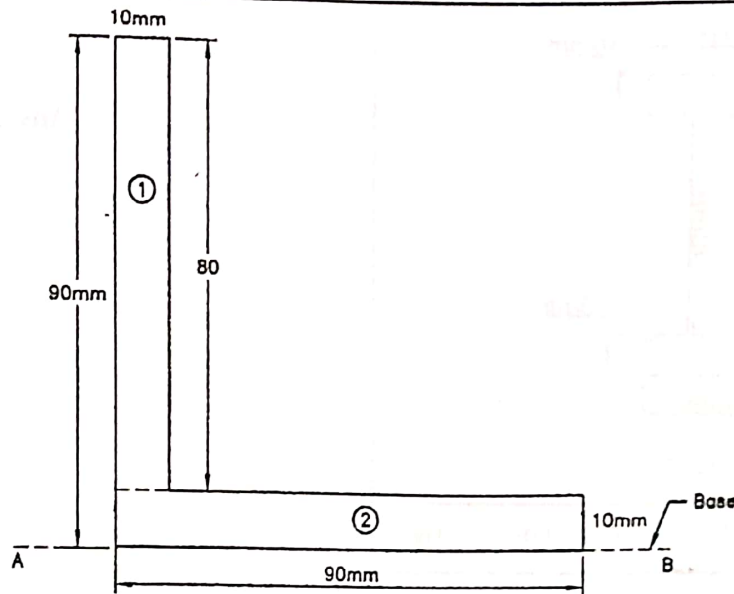


FIG. 3

[Ans. : $I_{\text{base}} = 2.456 \times 10^6 \text{ mm}^4$]

Ex. 4 Find M.I. about centroidal axes for I-section shown in fig.

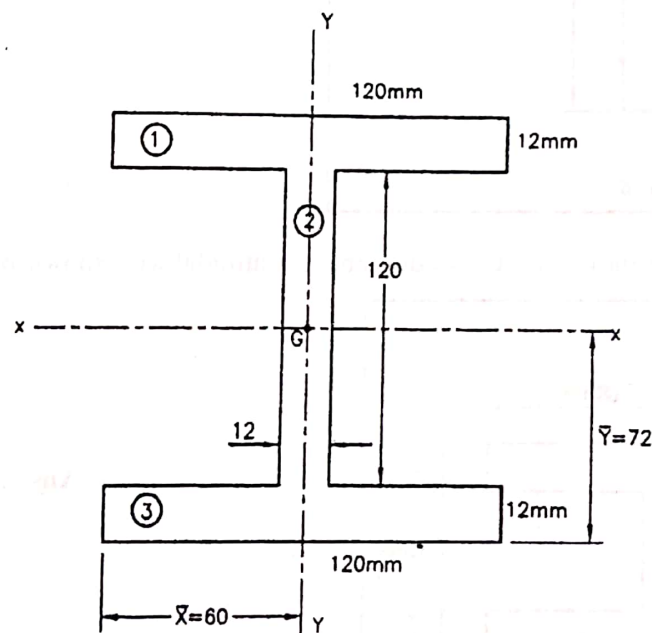


FIG. 4

[Ans. : $\bar{x} = 60 \text{ mm}$, $\bar{y} = 72 \text{ mm}$]

$I_{xx} = 14.306 \times 10^6 \text{ mm}^4$

$I_{yy} = 3.47 \times 10^6 \text{ mm}^4$]

Ex. 5 Find I_{xx} and I_{yy} for a channel section shown in fig.

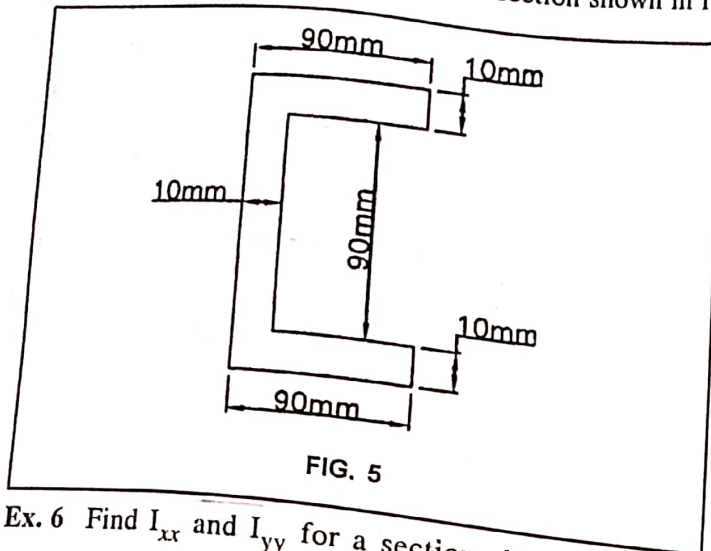


FIG. 5

[Ans. : $I_{xx} = 5.22 \times 10^6 \text{ mm}^4$
 $I_{yy} = 1.415 \times 10^6 \text{ mm}^4$]

Ex. 6 Find I_{xx} and I_{yy} for a section shown in figure.

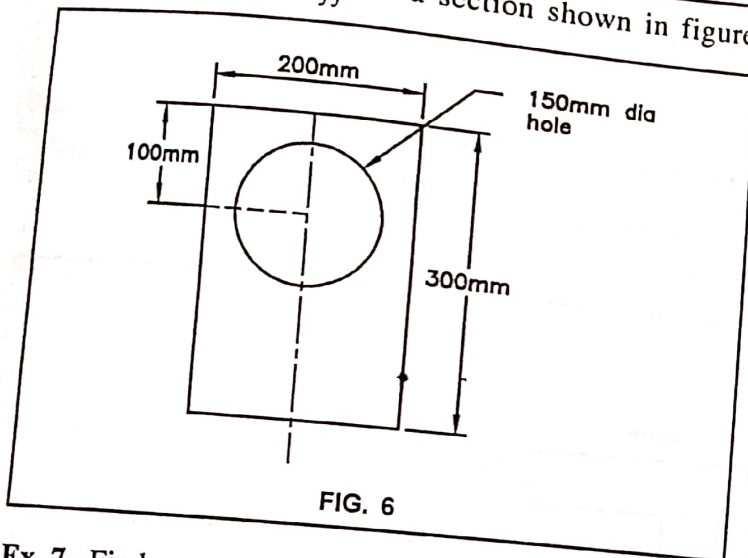


FIG. 6

[Ans. : $\bar{y} = 129.12 \text{ mm}$

$I_{xx} = 362.52 \times 10^6 \text{ mm}^4$

$I_{yy} = 175.15 \times 10^6 \text{ mm}^4$]

Ex. 7 Find moment of inertia of swastic section about centroidal axis shown below.

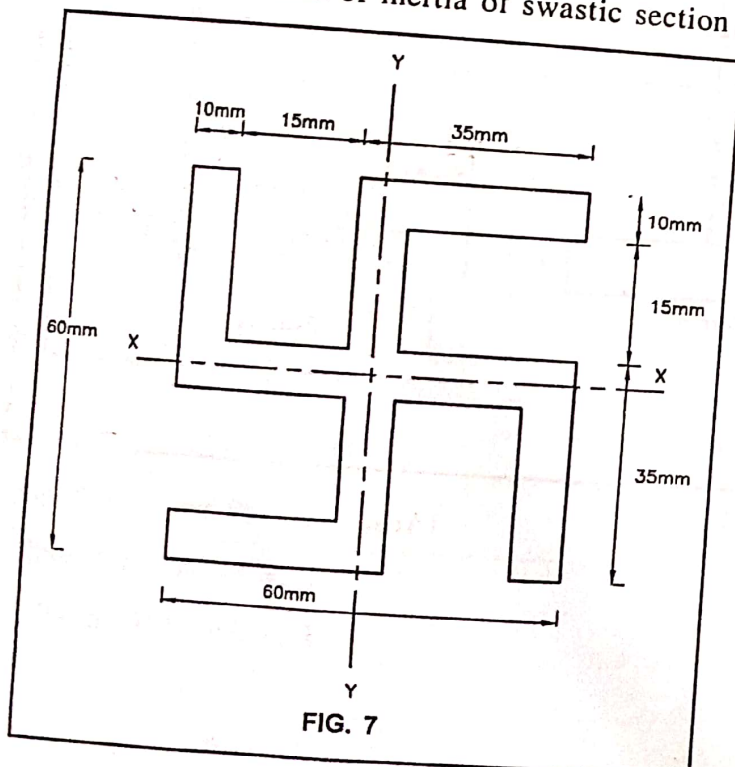


FIG. 7

[Ans. : $I_{xx} = I_{yy} = 0.68 \times 10^6 \text{ mm}^4$]

Ex. 8 Find M.I. about x axis and about the base of triangle shown below.

[Ans. : $I_{xx} = 1.97 \times 10^6 \text{ mm}^4$
 $I_{\text{base}} = 5.58 \times 10^6 \text{ mm}^4$]

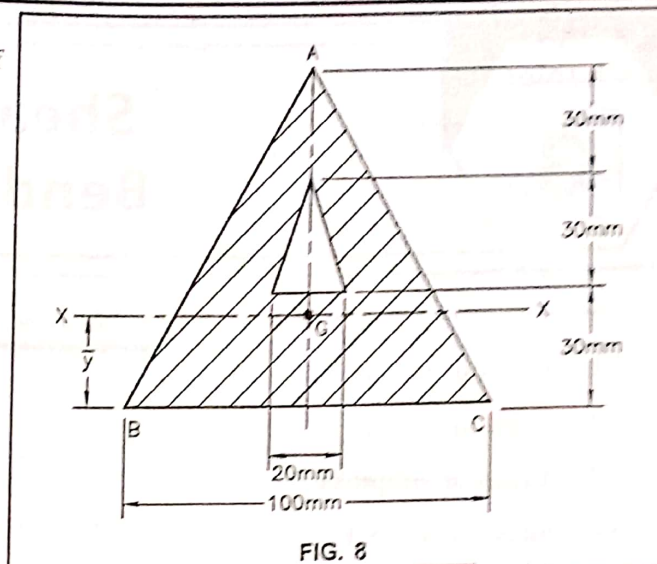


FIG. 8

Ex. 9 Find M.I. about $x-x$ axis for a given shaded area.

[Ans. : $\bar{y} = 154.4 \text{ mm}$

$I_{xx} = 3.22 \times 10^8 \text{ mm}^4$]

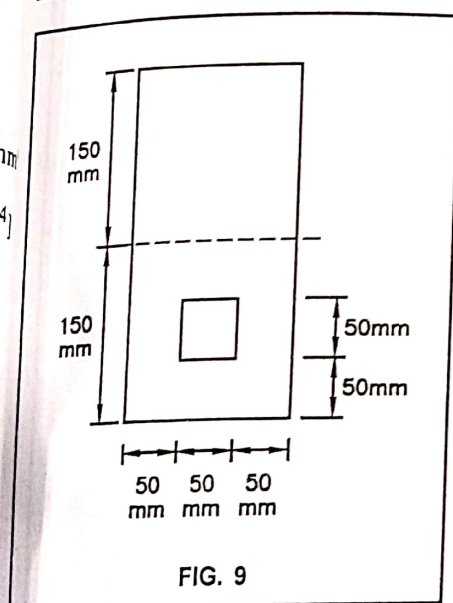


FIG. 9

Ex. 10 Find centroidal M.I. of a channel section shown below.

[Ans. : $I_{xx} = 27.5 \times 10^6 \text{ mm}^4$

$I_{yy} = 72.56 \times 10^6 \text{ mm}^4$]

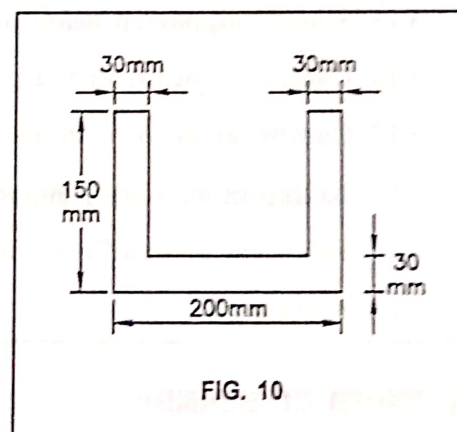


FIG. 10

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ch-4

EXERCISE

180 1. Write assumptions made in theory of bending.

(Nov. 2018, 2019, Feb. 2021, Sept. 2021, Feb. 2022)

181 2. State and explain equation of bending theory.

(Nov. 2017, 2018, 2019, May 2019, Feb. 2022)

3. Define :

181 (i) Neutral axis

(May 2017, Feb. 2021)

179 (ii) Bending stress

(May 2017, Feb. 2021)

181 (iii) Moment of resistance

(May 2015)

EXERCISE - NUMERICALS

Ex. 1 A rectangular section of beam $300 \text{ mm} \times 400 \text{ mm}$ is simply supported over a span 6 m. It is subjected to u.d.l. 120 kN/m over entire span. Find maximum bending stress.

[Ans. : $f = 67.5 \text{ N/mm}^2$]

Ex. 2 A cantilever beam 4 m span is subjected to u.d.l. 20 kN/m over entire span and a point load of 40 kN at free end. The section of beam is rectangular having depth equal to twice the width. If bending stress is not to exceed 100 N/mm^2 , decide dimensions of the beam.

[Ans. : 168.40 mm , $d = 336.81 \text{ mm}$]

Ex. 3 A steel plate having cross section $40 \text{ mm} \times 2 \text{ mm}$ is bent up at radius of 2 m. Find bending stress and moment of resistance of the plate.

[Ans. : $M = 2666 \text{ N.mm}$

$f = 100 \text{ N/mm}^2$]

Ex. 4 A hollow rectangular section of wood having outside dimension $300 \text{ mm} \times 600 \text{ mm}$ and internal dimension $150 \text{ mm} \times 300 \text{ mm}$. If bending stress in wood is not to exceed 50 N/mm^2 .

find : (a) Moment of resistance

(b) If beam is not provided with hole in the section, find M.R.

[Ans. : (a) $MR = 843.75 \times 10^6 \text{ N.mm}$

(b) $MR = 900 \times 10^6 \text{ N.mm}$]

Ex. 5 A wooden beam having $200 \times 300 \text{ mm}$ is simply supported, subjected to uniformly distributed load of 12 kN/m over entire span. If permissible bending stress is not to exceed 8 N/mm^2 , find span of the beam.

[Ans. : $l = 4000 \text{ mm}$]

☆☆☆

In this chapter we shall study the stresses set up in the beams due to bending only.

Q1 4.2 ASSUMPTIONS IN THE THEORY OF PURE BENDING :

(May 2017, Nov. 2017, 2018, 2019, Sept. 2021, Feb. 2021, 2022)

The following assumptions are made in the theory of pure bending :

1. The material of the beam is perfectly homogeneous (i.e. of the same kind throughout) and isotropic (i.e. of equal elastic properties in all directions).
2. The beam is stressed within its elastic limit and Hook's law is valid.
3. The transverse sections, which were plane before bending remain plane after bending.
4. Each layer of beam is free to expand or contract, independently.
5. The value of young's modulus (E) is the same in tension and compression.
6. Beam is straight before loading.
7. The cross-section of beam is uniform throughout its length.
8. The radius of curvature of beam is very large as compared to its c/s dimensions.

Moment of resistance, $M = f \times Z$.

4.4 EQUATION OF BENDING STRESS : (Flexure equation)

Q2 Consider a layer PQ at a distance y from the neutral axis.

After bending, layer PQ compressed to P'Q'.

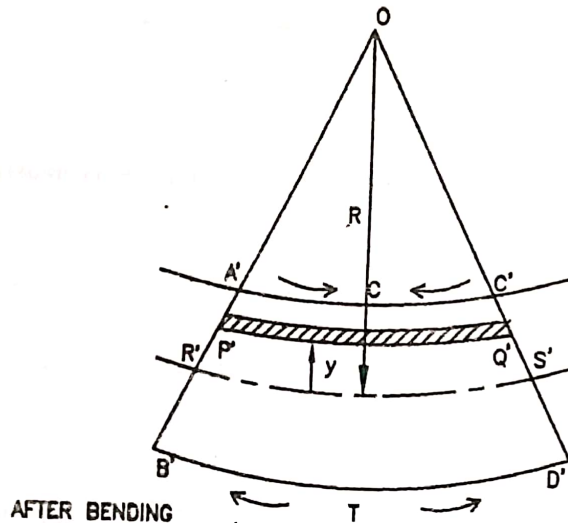
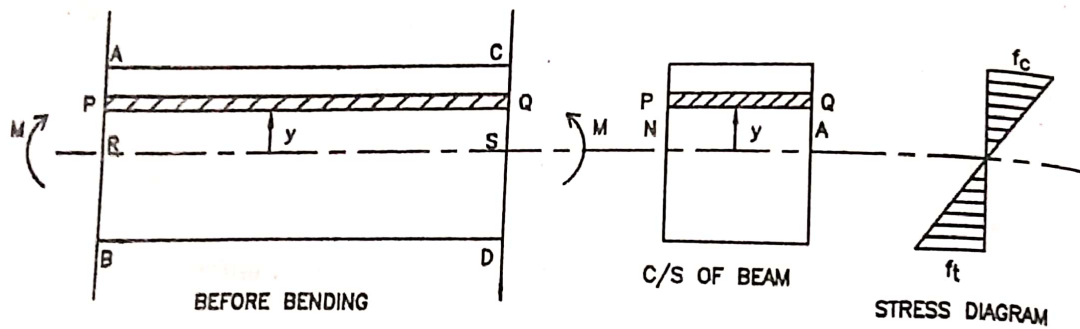


FIG. 4.3 BENDING THEORY

∴ Decrease in length of PQ layer,

$$\delta l = PQ - P'Q'$$

$$\therefore \text{Strain} = \epsilon = \frac{\delta l}{l} = \frac{PQ - P'Q'}{PQ} \dots\dots (i)$$

Now, from geometry of the figure, two sections $OP'Q'$ and $OR'S'$ are similar.

$$\frac{P'Q'}{R-y} = \frac{R'S'}{R}$$

$$\therefore \frac{P'Q'}{R'S'} = \frac{R-y}{R}$$

$$\therefore 1 - \frac{P'Q'}{R'S'} = 1 - \frac{(R-y)}{R}$$

$$\therefore \frac{R'S' - P'Q'}{R'S'} = \frac{R - R + y}{R}$$

$$\therefore \frac{R'S' - P'Q'}{R'S'} = \frac{y}{R} \quad (\because R'S' = RS = PQ = \text{Neutral layer})$$

$$\therefore \frac{PQ - P'Q'}{PQ} = \frac{y}{R} \dots\dots (ii)$$

$\therefore$ from equation (i) and (ii)

$$\epsilon = \frac{y}{R} \quad (\because \text{bending stress } (f) = \epsilon \cdot E)$$

$$\therefore \boxed{\frac{f}{E} = \frac{y}{R}} \quad \therefore \epsilon = \frac{f}{E}$$

$$\therefore \boxed{\frac{f}{y} = \frac{E}{R}} \dots\dots (iii)$$

Now, consider a small layer PQ of a beam section at a distance y from NA.

δa = Area of PQ layer

from equation (iii), $f = y \times \frac{E}{R}$

$\therefore$ total force in this layer,

$$= f \times \delta a$$

$$= y \times \frac{E}{R} \times \delta a$$

$\therefore$ Moment of this force about NA,

$$= \left(y \times \frac{E}{R} \times \delta a \right) \times y$$

$$= \frac{E}{R} \cdot y^2 \cdot \delta a$$

$\therefore$ Algebraic sum of all such moments about N.A.

$$M = \Sigma \frac{E}{R} \cdot y^2 \cdot \delta a$$

$$= \frac{E}{R} \Sigma y^2 \cdot \delta a$$

$$\text{but } \Sigma y^2 \cdot \delta a = I$$

$$\therefore M = \frac{E}{R} \cdot I$$

$$\therefore \boxed{\frac{M}{I} = \frac{E}{R}} \dots\dots (4)$$

$$\text{Curvature} = \frac{1}{R} = \frac{M}{EI}$$

$$\text{Strain} = \epsilon = \frac{f}{E} = \frac{y}{R}$$

is proportional to distance of fibre from N.A.

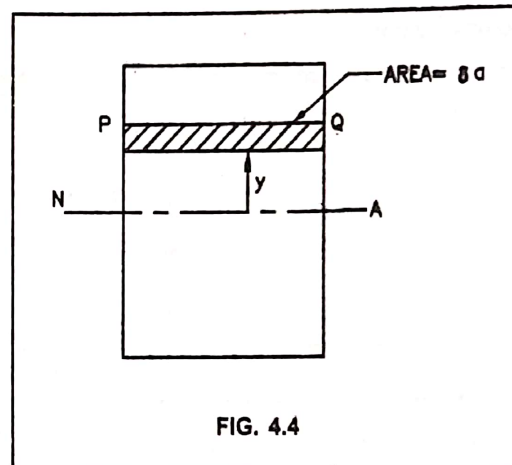


FIG. 4.4

∴ From equation (iii) and (iv),

$$\frac{M}{I} = \frac{f}{y} = \frac{E}{R}$$

... It is called bending equation or flexure equation.

(Nov.

Where, M = Moment of resistance

I = M.I. of the section about x-x axis

f = bending stress (maxi)

y = distance of extreme fibre from N.A. *very up and*

E = Modulus of elasticity

R = Radius of curvature of the beam

stretched. But, layer AB which is neither compressed nor stretched, is known as **neutral layer** or **neutral plane**.

- **Neutral Axis :** Q3(i)

(May 2017, Feb. 2021)

The line of intersection of the neutral layer, with any normal cross section of a beam is known as **neutral axis** of that section.

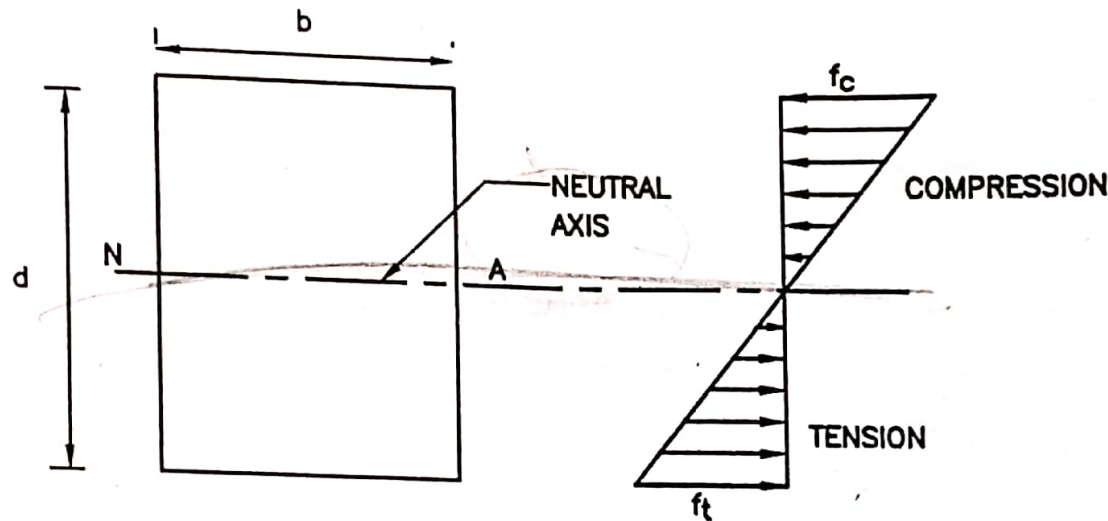


FIG. 4.2

If section of a beam is subjected to pure sagging bending moment, stresses above N.A. are compressive and stresses below N.A. are tensile. At the neutral axis (NA), there is no stress of any kind.

N.A. always passes through the centroid of the section of beam.

- **Moment of resistance :** Q3 (iii)

(May 2015)

When a beam length is subjected to a constant amount of bending moment, on one side of neutral axis there are compressive stresses and on the other there are tensile stresses. These stresses form a couple, whose moment must be equal to external moment M . The moment of this couple, which resists the external bending moment, is known as moment of resistance. $M = f \times Z$.

Pure bending stress :

Q3 (ii)

When a beam section is subjected to only bending moment (no shear, axial force, torsion), such a bending is called pure bending or simple bending and the bending stress produced is called pure bending stress.

For example, a beam subjected to equal and opposite couples at the ends, in the plane of bending as shown in Fig. 4.1 (a) is a case of pure bending. Similarly, the simply supported beam subjected to two equal point loads at C and D, creates a constant moment zone CD, free from shear as shown in Fig. 4.1 (b). Such a bending in zone CD is called pure bending or simple bending.

In pure bending beam bends into an arc of a circle.

In the portion of beam subjected to pure bending, $SF = 0$.

$$\therefore F = \frac{dM}{dx} = 0.$$

When bending is also accompanied by shear, it is called ordinary bending.

EXERCISE

Chapter 5

1. Explain the equation of shear stress $\tau = \frac{FA\bar{y}}{I.b}$. (May 2015, 2017, Feb. 2022)
2. Prove that for rectangular section $\tau_{\max} = 1.5 \tau_{\text{ave}}$. (Nov. 2018, Feb. 2021, Sept. 2021)
3. Derive relationship for maximum shear stress and average shear stress for circular cross section.
4. Draw shear stress distribution curve for following sections. (Nov. 2017, 2018, 2019, Sept. 2021, Feb. 2022)
 - (i) I-section
 - (ii) T-section
 - (iii) Rectangular section
 - (iv) Circular section
 - (v) L-section
 - (vi) C-section
 - (vii) Hollow rectangular section
 - (viii) Triangular section

EXERCISE - NUMERICALS

Ex. 1 A tee section shown in figure is subjected to a S.F. of 160 kN. Draw shear stress distribution diagram.
Take $I_{xx} = 9.6 \times 10^7 \text{ mm}^4$.

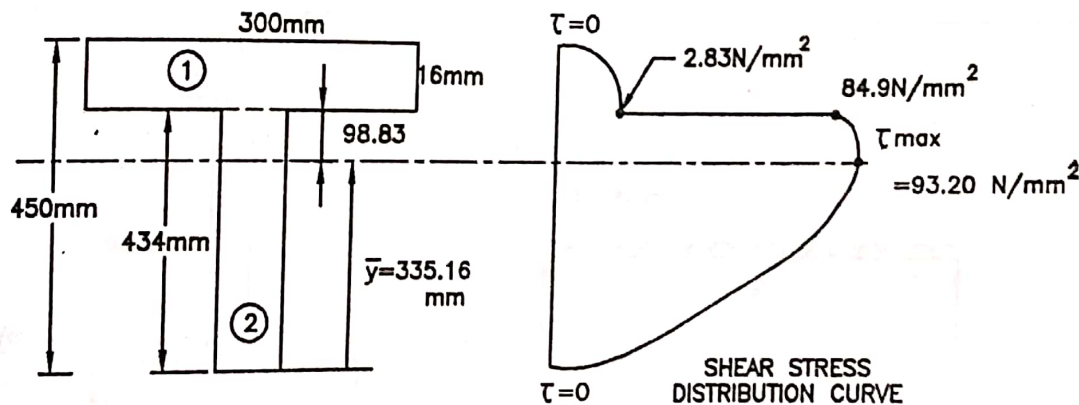


FIG. EX. 1

Ex. 2 An I-section having web $400 \times 12 \text{ mm}$ and flanges $300 \times 20 \text{ mm}$ is used as a simply supported beam to carry u.d.l. of 140 kN/m over entire span of 5.0 m . Draw shear stress distribution diagram for beam.
Take $I_{xx} = 6 \times 10^8 \text{ mm}^4$.

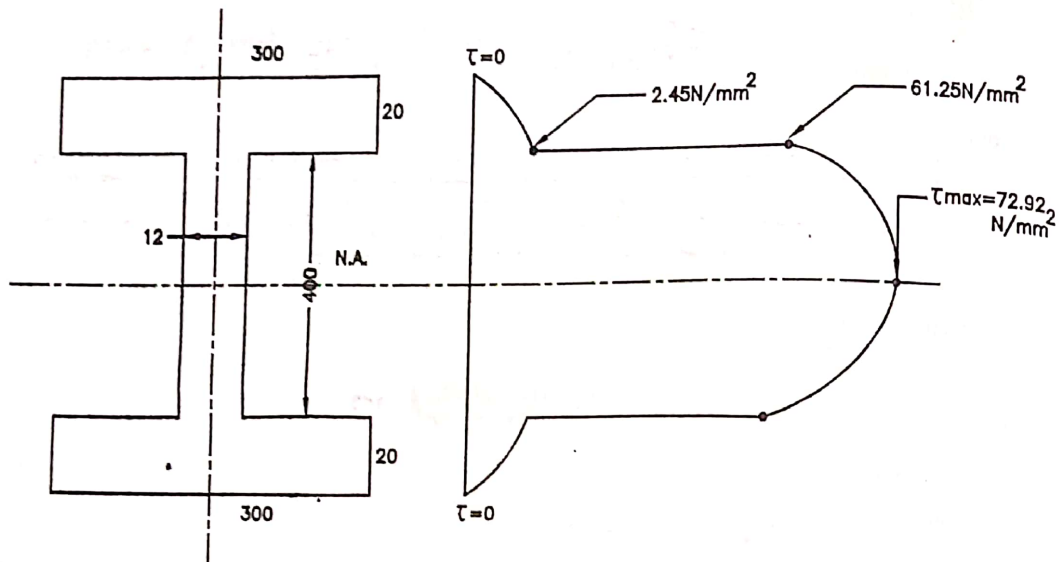


FIG. EX. 2

Ex. 3 A cantilever beam 3 m long is subjected to u.d.l. of 10 kN/m over entire span. The section of beam is 250 mm wide and 400 mm deep. Find average shear stress and maximum shear stress at the section of maximum shear force.

(Ans. : $\tau_{ave} = 0.3 \text{ N/mm}^2$

$\tau_{max} = 0.45 \text{ N/mm}^2$)

Ex. 4 A 50 mm $\times$ 100 mm in depth rectangular section of a beam is S/s at the ends with 2 m span. The beam is loaded with 20 kN point load at 0.5 m from right end. Calculate the maximum shearing stress.

(Ans. : $\tau_{max} = 4.5 \text{ N/mm}^2$)

Ex. 5 A simply supported beam is loaded as shown in figure. Draw shear stress distribution diagram across the section at the point of maximum shear force.

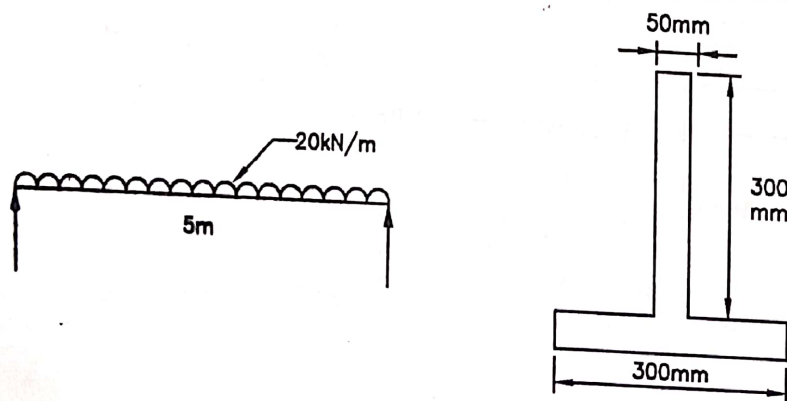


FIG. EX. 5

(Ans. : $I_{xx} = 345.3 \times 10^6 \text{ mm}^4$

$\tau_{max} = 4.08 \text{ N/mm}^2$

τ at junction = 0.633 & 3.8 N/mm^2)

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5.2 EQUATION FOR SHEAR STRESS IN A BEAM :

(May 2016, 2017, Feb. 2022)

Consider rectangular section of a beam shown below.

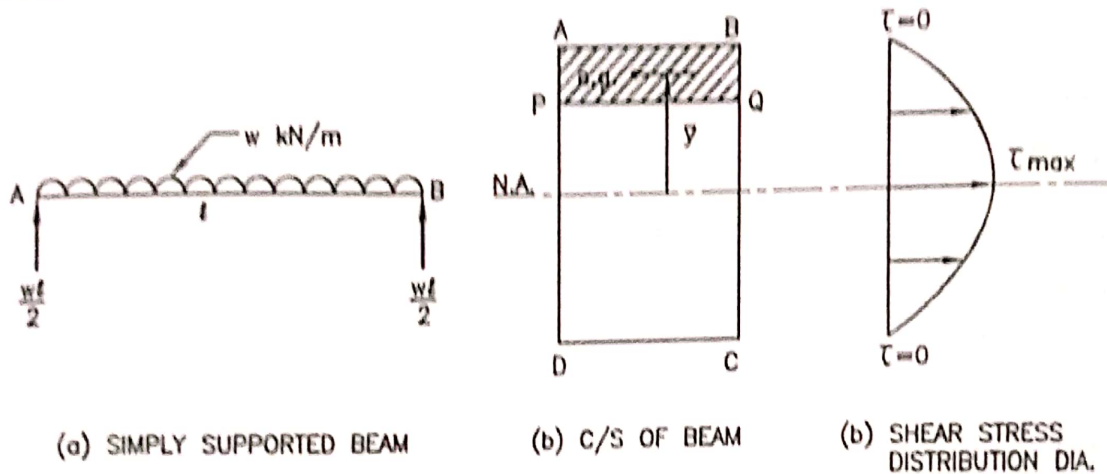


FIG. 5.1 (B)

The equation of shear stress is,

$$\tau = \frac{F \cdot A \cdot \bar{y}}{I \cdot b}$$

Where, τ = shear stress at given layer (PQ)

F = shear force

A = Area of the section above or below the given layer (PQ)

$\bar{y}$ = distance of c.g. of the area from N.A.

I = M.I. of the section about N.A.

b = Width of the section at given layer.

• Average shear stress :

$$\text{Average shear stress} = \frac{\text{Shear force}}{\text{Shear Area}}$$

$$\tau_{ave} = \frac{F}{A} \dots \text{N/mm}^2$$

5.3 SHEAR STRESS DISTRIBUTION FOR RECTANGULAR SECTION : Q2

(Nov. 2018, Feb. 2021, Sept. 2021)

Consider a rectangular section of width 'b' and depth 'd' as shown in figure
Maximum shear stress will occur at N.A.

$\therefore A = \text{Area of section above N.A.}$

$$= b \times \frac{d}{2}$$

$\bar{y} = \text{distance of c.g. of area above N.A.}$

$$= \left(\frac{d}{2} \right) = \frac{d}{4}$$

$$I = \frac{bd^3}{12}$$

Now,

Shear stress at neutral axis,

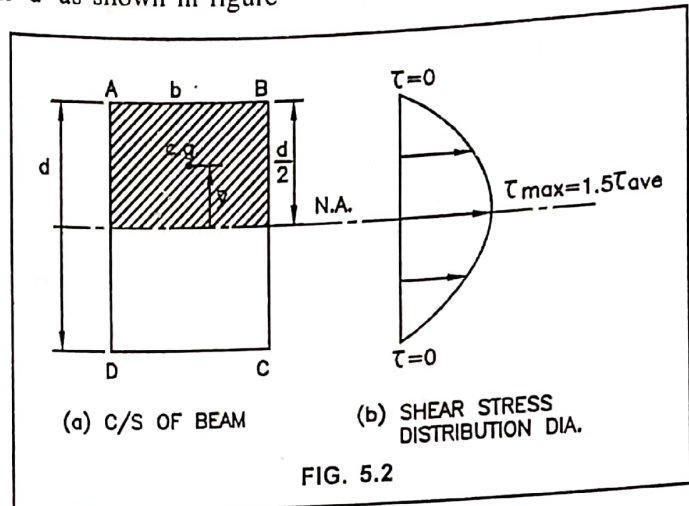
$$\tau = \frac{FA \cdot \bar{y}}{I \cdot b}$$

$$\therefore \tau_{max} = \frac{F \times b \times \frac{d}{2} \times \frac{d}{4}}{\frac{bd^3}{12} \times b}$$

$$= \frac{3}{2} \times \frac{F}{b \cdot d}$$

$$= 1.5 \frac{F}{A}$$

$$\therefore \tau_{max} = 1.5 \tau_{ave}$$



$$\therefore \tau_{ave} = \frac{F}{A}$$

5.4 SHEAR STRESS DISTRIBUTION FOR CIRCULAR SECTION : Q3

Consider a circular section of diameter 'd' as shown in figure.

Maximum shear stress will occur at N.A.

$\therefore A = \text{Area of section above N.A.}$

$$= \frac{\pi d^2}{2} = \frac{\pi d^2}{8}$$

5.5 SHEAR STRESS DISTRIBUTION DIAGRAMS FOR DIFFERENT CROSS- SECTIONS OF BEAM :

(May 2019, Nov. 2019, Feb. 2021, 2022, Sept. 2021)

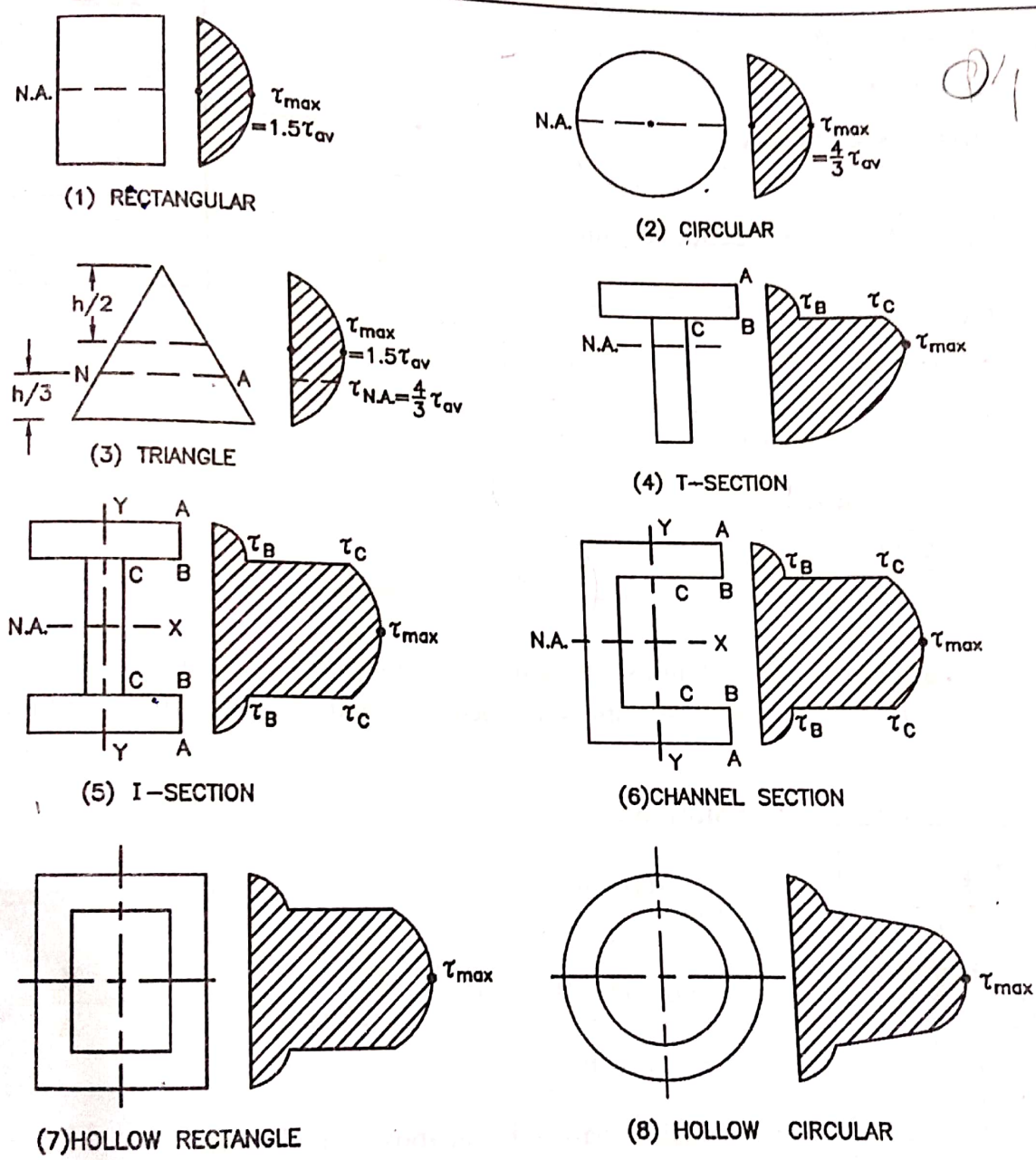


FIG. 5.4 SHEAR STRESS DISTRIBUTION

EXERCISE

ch-8

1. Define :

(i) Column

(ii) Crushing load

(iv) Slenderness ratio

(v) Radius of gyration

(iii) Buckling load

(May 2017)

2. Explain different end conditions of column and effective length with sketches.

(May 2017, 2019, Nov. 2017, 2019, Feb. 2022)

3. Explain limitations of Euler's formula.

4. Differentiate between column and strut.

(May 2015)

5. Give assumptions of Euler's theory.

(May 2016, 2017, Nov. 2018, 2019)

EXERCISE - NUMERICALS

Ex. 1 : A hollow circular column 6 m long having 100 mm internal diameter and 10 mm thickness is fixed at both ends. $E = 2 \times 10^5 \text{ N/mm}^2$. Calculate Euler's buckling load.

(Ans. : $P_E = 1155.85 \text{ kN}$)

Ex. 2 : A steel rod 5 m long and 40 mm diameter is used as a column with one end fixed and other end free. Determine crippling load by Euler's formula. $E = 200 \text{ kN/mm}^2$.

(Ans. : $P_E = 2.48 \text{ kN}$)

- Ex. 3 : A hollow rectangular column 6 m long has outer dimension 60 mm $\times$ 80 mm and 6 mm thickness, with one end hinged and other fixed. Calculate Euler's buckling load. $E = 200$ GPa. [Ans. : $P_E = 89.18$ kN]
- Ex. 4 : A hollow circular C.I. column has external diameter 300 mm and 40 mm thickness, 8 m length. Determine Euler's crippling load. [Ans. : $P_E = 17433.42$ kN]
 $E = 100$ kN/mm². Column is fixed at both ends.
- Ex. 5 : A hollow circular column has external diameter 100 mm and 25 mm thickness. The both ends of column are fixed. If Euler's load is 505 kN, $E = 100$ GPa, find length of the column. [Ans. : $l = 6$ m]
- Ex. 6 : A steel tube having external diameter 300 mm and internal diameter 250 mm is fixed at both ends. Length of column is 5.0 m and F.O.S. is 4. Calculate safe load by Rankine formula, $\alpha = \frac{1}{1600}$
 $f_c = 600$ N/mm² [Ans. : $P_R = 2298$ kN]
- Ex. 7 : A steel tube, 4 m long is used as a column with both ends fixed. The external diameter of tube is 50 mm and thickness 10 mm. $E = 2 \times 10^5$ N/mm², $f_c = 320$ N/mm².
 Calculate : (a) Buckling load by Euler's formula
 (b) Crippling load by Rankine formula
- Rankine constant $\alpha = \frac{1}{7500}$ [Ans. : $P_E = 131.76$ kN, $P_R = 114.56$ kN]
- Ex. 8 : A steel tube with 60 mm external diameter and 10 mm thickness is used as a column. If slenderness ratio = 90, calculate load carrying capacity of column.
 $f_c = 320$ N/mm, $\alpha = \frac{1}{4800}$
 Column has one end fixed and other hinged.
 Also find length of column. [Ans. : $l = 2294$ mm]
- Ex. 9 : A 'T' section has a flange 180 $\times$ 15 mm and web 235 $\times$ 10 mm is used as a column, with both ends fixed. Using Rankine formula calculate safe load.
 $f_c = 335$ N/mm²
 $\alpha = \frac{1}{7500}$
 F.O.S. = 3 [Ans. : $P_{\text{safe}} = 439.83$ kN]
- Ex. 10 : A hollow C.I. circular section is used as a column with 4 m length and both ends fixed. It is carrying axial load of 250 kN. Using F.O.S. = 5 calculate size of column. Take internal diameter of column = 0.8 external diameter.
 $f_c = 550$ N/mm²
 $\alpha = \frac{1}{1600}$ [Ans. : $D = 136.3$ mm, $d = 109.04$ mm]

☆☆☆

8.2 COLUMN : Q 1 (i)

When strut is vertical it is known as column.

- The cross - sectional dimensions of column are large.
- Normally, columns carry heavy compressive loads.
- Columns are used in concrete and steel buildings.

8.7 **CRUSHING LOAD :** (i) 1 (ii)

In case of short columns, with increase in axial compressive load, compressive stress increases. After some load, the column fails by crushing.

The load at which, short column fails by crushing is called **crushing load**.

The stress corresponding to crushing load is called **crushing stress**.

8.8 CRIPPLING LOAD OR BUCKLING LOAD OR CRITICAL LOAD : 01 (iii)

(May 2017)

In case of long columns, with increase in axial compressive load, compressive stress increases. After some load, the column starts buckling (bending) and bending stress also produces. Finally, the column fails by buckling.

The load at which, long column starts buckling is called **buckling load or crippling load**.

For long columns, direct stress (σ_o) is negligible as compared to bending stress (σ_b).

A column tends to buckle about an axis of least moment of inertia called minor axis.

• Buckling of column depends upon the following factors.

1. Amount of load
2. Length of column
3. End conditions of column
4. Cross - sectional dimensions of column
5. Material of column.
6. Radius of gyration.

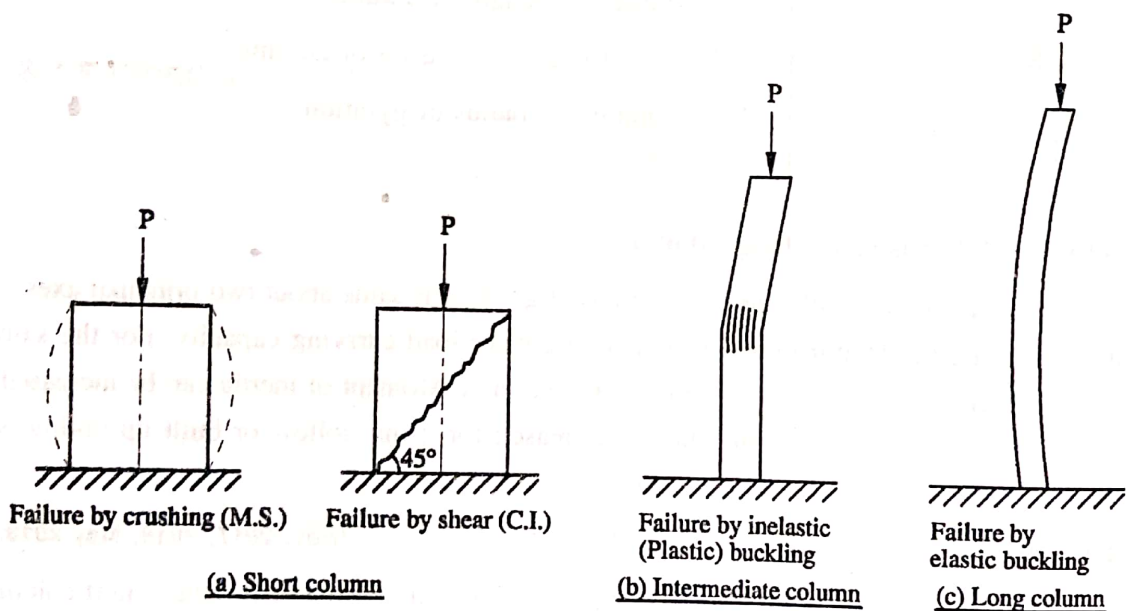


FIG. 8.2 (A) FAILURE OF COLUMNS

SLENDerness RATIO (λ) :

Q 1 (iv)

(June 2014, May 2017)

$$\text{Slenderness ratio} = \frac{\text{Effective length of column}}{\text{Minimum radius of gyration}}$$

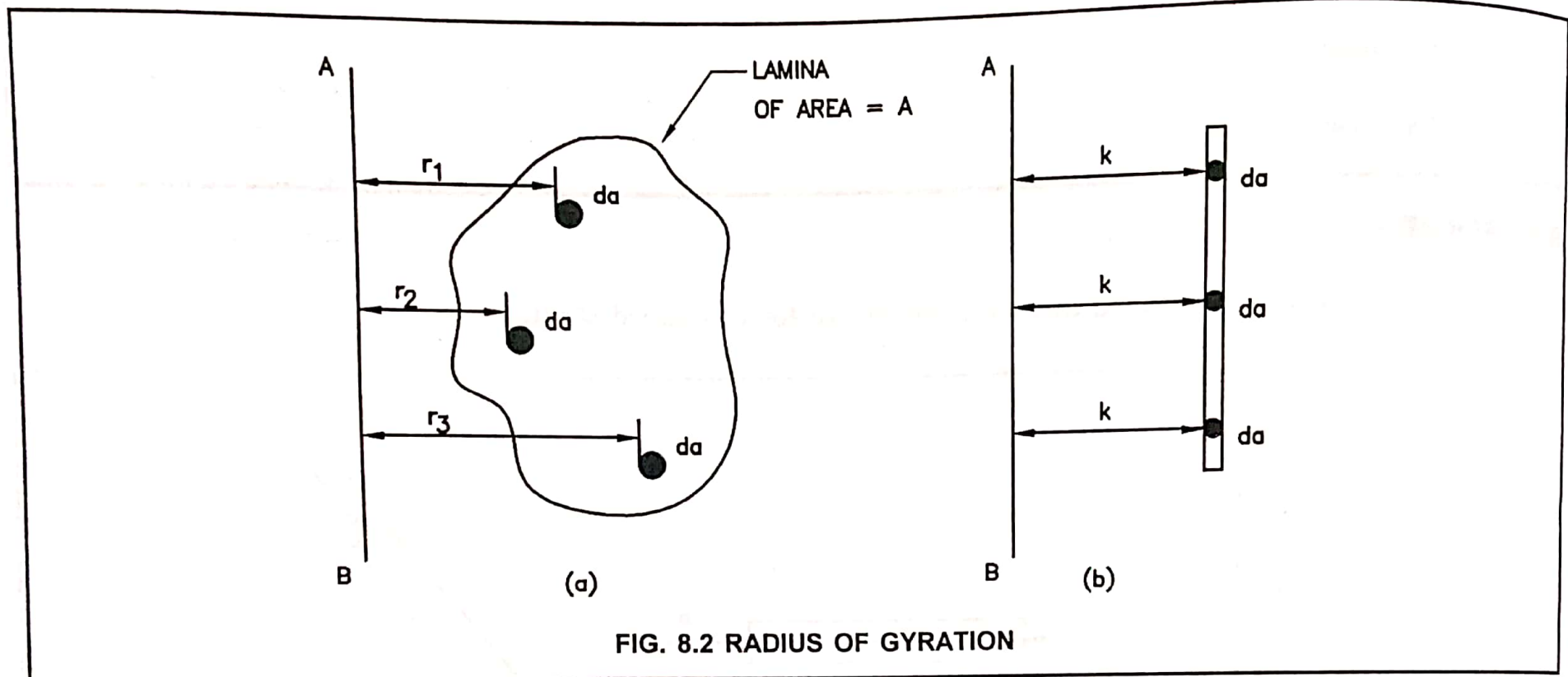
$$\therefore \lambda = \frac{l_e}{k_{\min}}$$

- If λ for column is more, its load carrying capacity will be less.
- If λ for column is less, its load carrying capacity will be more.



8.3 RADIUS OF GYRATION (k) : Q1 (V)

The distance from the given axis at which, if all the small elements of the lamina are placed, the M.I. of the lamina about the given axis does not change. This distance is called **Radius of gyration**.



Mathematically,

$$k = \sqrt{\frac{I}{A}} \quad \text{or} \quad I = A k^2$$

8.9 COLUMN END CONDITIONS AND EFFECTIVE LENGTH :

Q2

(May 2019, Nov. 2019, Feb. 2021, Sept. 2021, Feb. 2022)

We may consider the following four end conditions.

1. Both ends hinged.
2. Both ends fixed
3. One end fixed and other hinged.
4. One end fixed and other free.

(i) **Both ends hinged :** (Fig. 8.3)

$$l_e = l$$

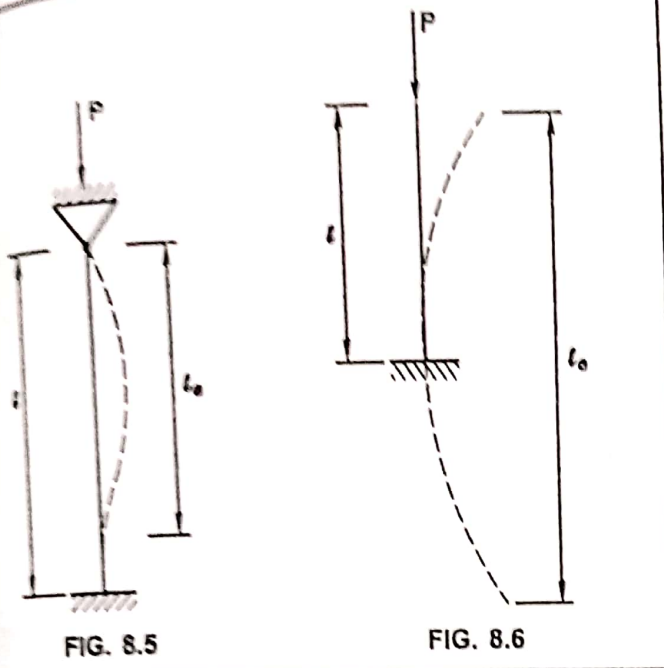
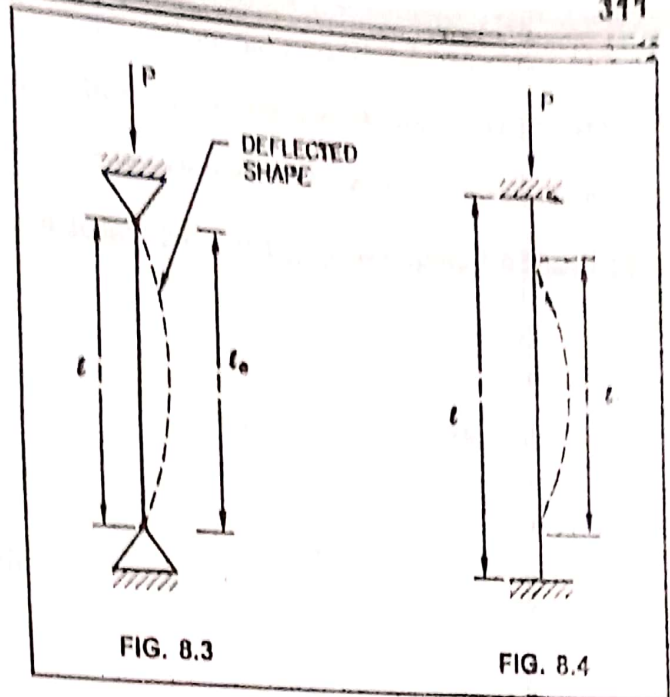
where,

l_e = effective length of column

l = actual length of column

(ii) **Both ends fixed :** (Fig. 8.4)

$$l_e = \frac{l}{2}$$



(iii) **One end fixed and other hinged :** (Fig 8.5)

$$l_e = \frac{l}{\sqrt{2}}$$

(iv) **One end fixed and other free :** (Fig 8.6),

$$l_e = 2l$$

8.10 EULER'S FORMULA FOR CRIPPLING LOAD :

$$P_E = \frac{\pi^2 EI}{(l_e)^2}$$

Where, P_E = Euler's crippling load

E = Modulus of elasticity

I = M.I. (Minimum)

l_e = Effective length of column

σ. The shortening of column due to axial compression

(May 2018)

8.12 LIMITATIONS OF EULER'S FORMULA :

The Euler's formula is,

$$P_E = \frac{\pi^2 EI}{(l_e)^2}$$

$$\therefore P = \frac{\pi^2 \cdot E \cdot Ak^2}{(l_e)^2}$$

$$\frac{P}{A} = \frac{\pi^2 E}{\left(\frac{l_e}{k}\right)^2}$$

$$\therefore \sigma = \frac{\pi^2 E}{(\lambda)^2}$$

$$\therefore \lambda^2 = \frac{\pi^2 E}{\sigma}$$

Substitute $I = Ak^2$

$$\therefore \sigma = \frac{P}{A} = \text{Crippling stress}$$

$$\frac{l_e}{k} = \lambda$$

The crippling stress depends upon the value of E and λ^2

When the value of λ is more, σ will be less

For mild steel,

$$\sigma = 330 \text{ N/mm}^2$$

$$E = 2.1 \times 10^5 \text{ N/mm}^2$$

$$\therefore \lambda^2 = \frac{\pi^2 \times 2.1 \times 10^5}{330}$$

$$\therefore \lambda^2 = \frac{\pi^2 \times 2.1 \times 10^5}{330}$$

$$\therefore \lambda^2 = 6280.65$$

$$\therefore \lambda = 79.25 \text{ say } 80.$$

Hence, when slenderness ratio (λ) for column is less than 80, the column is considered to be short and Euler's formula cannot be applied.

The Euler's formula is applicable to the long columns having $\lambda \geq 80$

are used in concrete and steel buildings.

• **Difference between strut and column :**

(May 2016, Nov. 2018, Sept. 2020)

Strut	Column
(1) It carry light axial compressive force.	(1) It carry heavy compressive force.
(2) It may be vertical, horizontal or inclined.	(2) It is always vertical.
(3) The c/s dimensions of strut are small.	(3) The c/s dimensions of column are large.
(4) It is used in roof trusses and bridge truss.	(4) It is used in concrete and steel buildings.

(May 2016, 2017, Nov. 2020)



8.11 ASSUMPTIONS OF EULER'S FORMULA :

1. The material of column is elastic, homogeneous and isotropic.
2. The column is long.
3. The load on column is truly axial.
4. Failure of column is due to buckling.
5. The cross section of column is uniform throughout its length.

6. The Hooke's law is valid.
7. The column is straight before application of load.
8. The shortening of column due to axial compressive load is neglected.